Big Data Final Report

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1 Introduction

The goal of this project is to design a prediction model that calculates the chances of an NBA team to win the championship using statistics from the regular season. The standard evaluation of NBA teams usually involves very basic statistics like victories and defeats, or playoff seeding. However, they do not fully reflect how powerful a team is. Two teams with identical statistics might be quite different in terms of efficiency and other parameters of playing. That is why probability-based modeling proves to be more accurate and reliable for predicting the chances to win the championship.

The current research considers the task of predicting the champion as a supervised machine learning problem where past results of the teams are considered to estimate the probability of winning the championship. In contrast to the creation of a straightforward classification model that would predict whether the team wins the championship or does not, this study utilizes probability estimates that consider the chances of teams.

In addition to predicting championship probabilities, this project also serves as an application of high-dimensional statistical learning techniques. With many explanatory variables being used in the study together with some interaction terms, the model becomes complicated and may require regularization techniques for proper functioning without being prone to overfitting. This problem is addressed by the use of LASSO regularized logistic regression in the project. This approach ensures the prediction of championship probabilities alongside variable selection.

2 Literature Review

Sports analytics has increasingly relied on statistical learning methods to evaluate team and player performance. Early basketball analytics research focused heavily on efficiency-based metrics and possession-level analysis. Dean Oliver introduced many foundational concepts in basketball analytics in Basketball on Paper, emphasizing the importance of offensive efficiency, defensive efficiency, rebounding, and turnovers in determining team success. His work demonstrated that efficiency-based metrics are often more predictive than traditional counting statistics.

As sports datasets became larger and more complex, machine learning and regularization techniques became more common in predictive sports modeling. Robert Tibshirani introduced the LASSO method, which applies coefficient shrinkage and variable selection simultaneously. LASSO is particularly useful in high-dimensional settings where predictors may be highly correlated or where interaction terms significantly increase the number of variables. The ability of LASSO to reduce overfitting while improving interpretability makes it relevant for the current project because the engineered interaction terms create a high-dimensional feature space with correlated predictors.

Modern statistical learning methods are discussed extensively by Trevor Hastie, Robert Tibshirani, and Jerome Friedman in The Elements of Statistical Learning. Their work highlights how regularization methods improve predictive performance while controlling model complexity in high-dimensional data environments.

Basketball analytics has also adopted regression-based frameworks to estimate player and team impact. Regularized Adjusted Plus-Minus, commonly referred to as RAPM, was developed by Joseph Sill and applies regularization techniques to basketball data to improve stability and reduce noise in player evaluation models. Although RAPM focuses primarily on player-level impact, its methodology demonstrates how regularization can effectively handle noisy and highly correlated basketball data. This project builds upon these ideas by applying regularized statistical learning methods to team-level championship prediction.

3 Data

The dataset for this study was acquired from an NBA API which has information on NBA teams statistics. The data for 2000-2025 seasons was chosen to be used as the training set to predict the 2026 season. Every instance in the dataset constitutes one NBA team's performance in a certain season, leading to about 800 observations of team-seasons. Using more than one season enhances the amount of observations and makes it possible to recognize broad patterns of history related to winning the title.

The response variable is a binary indicator representing championship outcome. Teams that won the NBA championship were labeled as one, while all other teams were labeled as zero. Although the response variable is binary, the primary goal of the project is not simple classification. Instead, the model estimates the probability that a team wins the championship given its regular season characteristics.

There are numerous metrics that represent each team's performance from the dataset, namely Wins, Losses, Offensive Efficiency, Defensive Efficiency, Pace, and Simple Rating System figures. Furthermore, additional features were created through engineering, thus increasing the number of predictor variables to include. Net Efficiency, defined as the difference between Offensive Efficiency and Defensive Efficiency, Win Percentage, and various interaction variables involving Offensive and Defensive Efficiency measures were created. Interaction variables increase the complexity of the dataset while making the data more appropriate for applying regularized regression models, with about 40-60 predictors to work with.

A number of preprocessing operations were done before modeling. Rows containing average data for the league were dropped from the dataframe, any categorical variables were changed to numeric where applicable, and interaction variables were engineered. Due to the different scales at which most basketball statistics operate, standardization was performed on all predictors before modeling. There was also a problem of class imbalance because of the nature of the dataset, wherein there is just one champion each year. To address this imbalance, balanced class weighting was implemented during model training.

4 Methodology

This project uses LASSO-regularized logistic regression to estimate championship probabilities. Logistic regression was selected because the response variable is binary and the model naturally produces probability estimates between zero and one. The model estimates the probability of a team winning the championship based on a set of predictor variables representing team performance characteristics.

Given that the data set has various correlated features, as well as created interactions, it made sense to use some sort of regularization technique to ensure that there was no over-fitting. LASSO regularization adds a penalty term to the objective function for logistic regression. The LASSO regularization term reduces irrelevant feature weights to zero and ensures that only the relevant features are selected during training.

The implementation was completed in Python using pandas for data manipulation and scikit-learn for modeling. The modeling pipeline included data collection, cleaning, feature engineering, train-test splitting, feature standardization, model fitting using cross-validated LASSO logistic regression, and probability prediction. The model was evaluated using log-loss and ROC AUC, which are more appropriate than simple accuracy because the primary focus of the project is probability estimation rather than binary classification.

LASSO logistic regression model was applied on the training data (2000-2025 seasons) and subsequently used to generate predictions for the current 2026 season. The output was normalized such that the percentage adds up to 100 for the 30 teams in the league. The normalization step was used to improve interpretability as championship odds across teams, similar to sportsbook-style implied probabilities.

5 Output

The main result of this study is a set of probabilities for each NBA team to win the championship according to the statistics from their regular season games. The algorithm does not predict just the winning or losing outcome of a game but rather produces the probability of winning the championship, which gives an additional level of meaning to the results of its operation.

The results suggest that efficiency-based metrics and team record were the most important predictors of championship success in the model. Variables such as effective field goal percentage, true shooting percentage, and win percentage are expected to play major roles in the model because championship teams typically are elite at scoring the basketball and win games during the regular season. The model achieved an ROC AUC score of 0.95 and a log-loss value of 0.28 on the testing set, suggesting strong ranking performance and well-calibrated probability estimates.

At first glance at the results, it is fair to say that the predicted probabilities are somewhat reasonable. The top teams from the model are teams that are in the playoffs and are considered by many fans and analysts to be championship contenders. The bottom teams are teams that are outside of the playoffs and projected to have a high lottery pick in the next season. As of mid May 2026, the second round of the playoffs is about to conclude, and the four remaining teams in the playoffs are the top 6 teams in the model. When we also look at past seasons in our training data, every team since 2000 ranked top 5 in probability, and all of these teams had an average rank of 2.2 in the model.

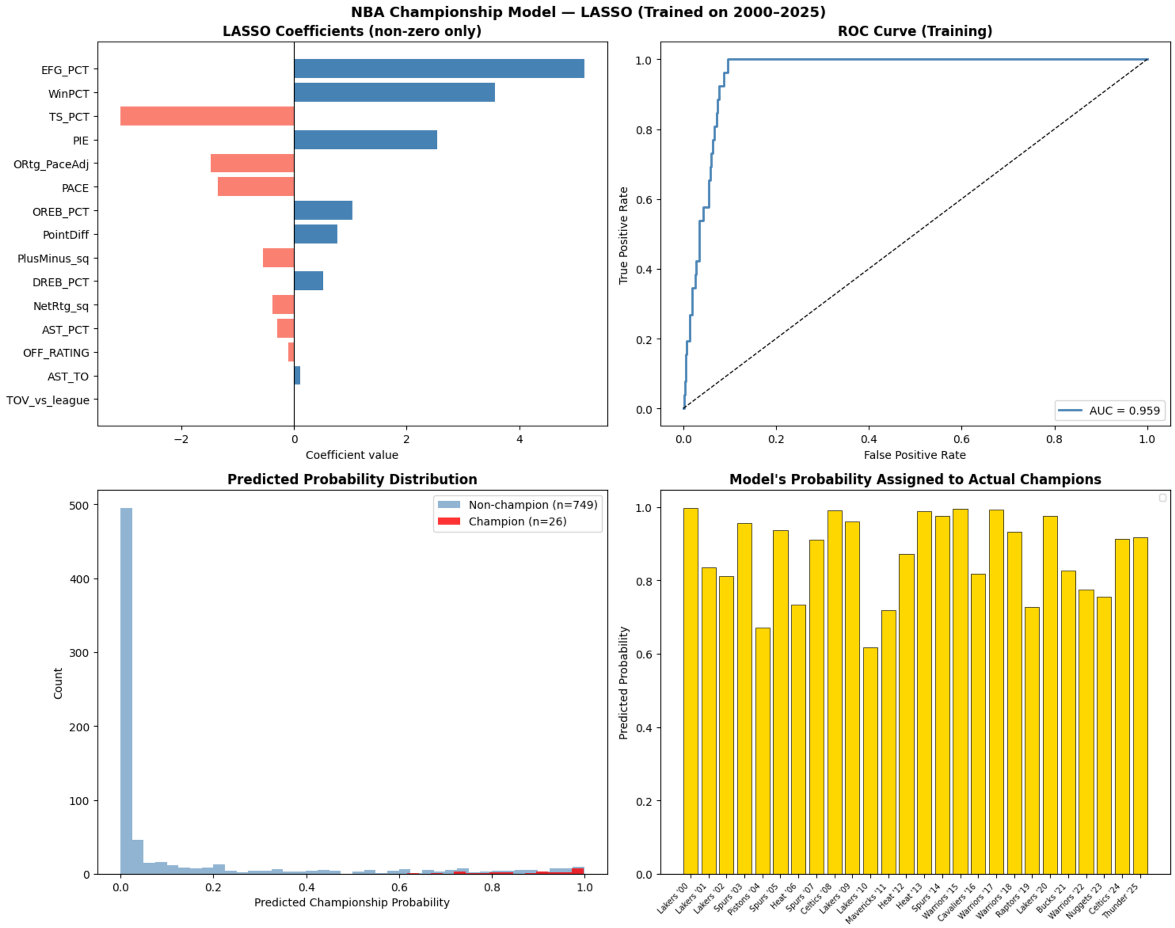

Figure 1: Model Evaluations

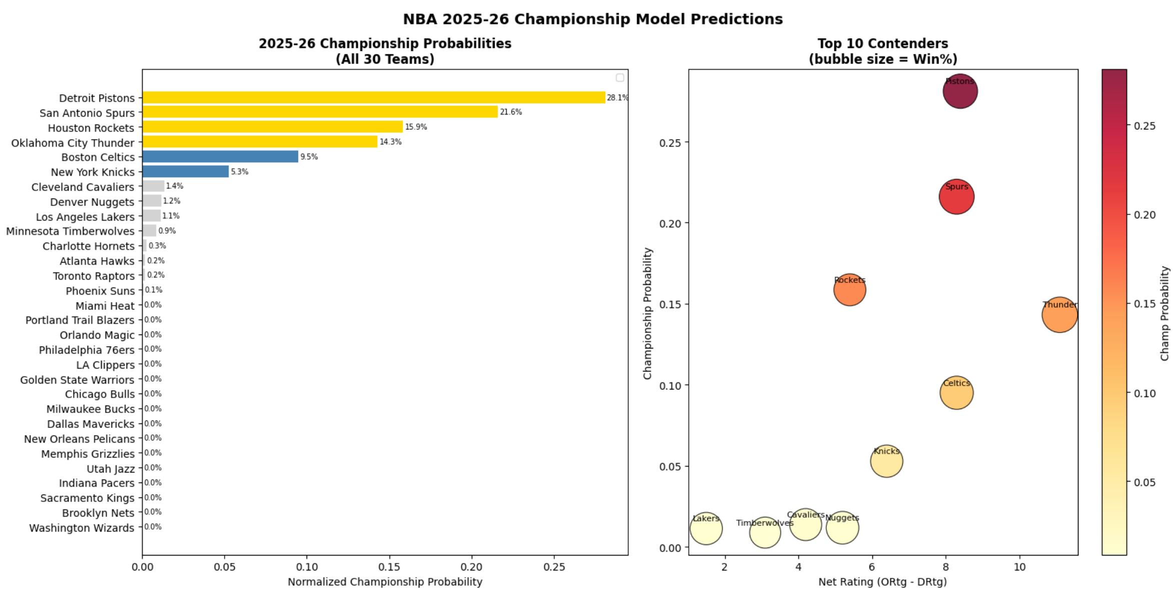

Figure 2: Championship Predicted Probabilities

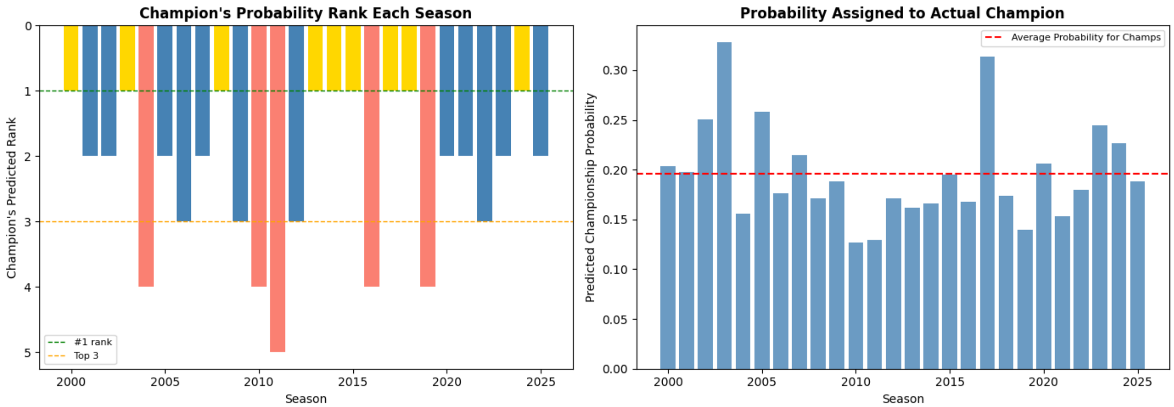

Figure 3: Historical Validation

References

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Joseph Sill (2010). Improved NBA Adjusted Plus-Minus Using Regularization.
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